

EC400

Lecture 2

Statistics I

Probability and Statistical Inference ■ Tom Glinnan

Plan for today

- Events, random variables, CDFs and densities.
- Moments and quantiles: summarising a distribution.
- Uniform and normal distributions; transformations.
- Joint and conditional distributions; Bayes' rule.
- Conditional moments, iterated expectations and total variance.
- Independence, mean independence and zero covariance; links to OLS.

Appendix non-examinable

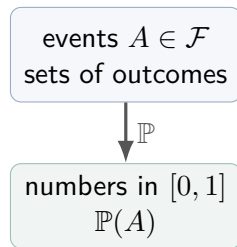
Defining expectation with integrals; proof of total variance.

Distributions

Probability Theory

- The **sample space** Ω contains every possible outcome.
- An **event** A is a set of outcomes, drawn from a collection $\mathcal{F}$ (sigma algebra). Its **complement** $A^c = \Omega \setminus A$ is the event that A does not occur.
- A probability function assigns a number to each event:

$$\mathbb{P} : \mathcal{F} \longrightarrow [0, 1], \quad A \longmapsto \mathbb{P}(A).$$



The three probability rules

A probability map obeys three basic rules:

- Total mass:

$$\mathbb{P}(\Omega) = 1.$$

- Non-negativity: $\mathbb{P}(A) \geq 0$ for every event A .
- Addition for disjoint events: if the A_j cannot occur together, then

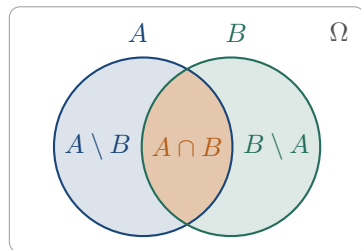
$$\mathbb{P}\left(\bigcup_{j=1}^{\infty} A_j\right) = \sum_{j=1}^{\infty} \mathbb{P}(A_j).$$

These rules imply $\mathbb{P}(\emptyset) = 0$ and $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.

The union bound

Two events: split the union into disjoint pieces.

$$\begin{aligned}\mathbb{P}(A \cup B) &= \mathbb{P}(A) + \mathbb{P}(B \setminus A) \\ &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) \\ &\leq \mathbb{P}(A) + \mathbb{P}(B).\end{aligned}$$



The inequality uses **non-negativity**: $\mathbb{P}(A \cap B) \geq 0$.

By induction...

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n \mathbb{P}(A_i).$$

The events may overlap; independence is **not** required.

Random variables

- Formally, a **random variable** is a (measurable) function

$$X : \Omega \rightarrow \mathbb{R}.$$

- Its **distribution** is the ‘pushforward measure’

$$\mu_X(A) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \in A\}), \quad \text{for an event } A.$$

e.g. $\mathbb{P}(X = 5)$ is the probability of the ω for which $X(\omega) = 5$.

The sum of two dice

Roll two fair six-sided dice. The sample space is

$$\Omega = \{(d_1, d_2) : d_1, d_2 \in \{1, \dots, 6\}\}.$$

Each outcome records both dice; for example,

$$\omega = (1, 4), \quad \omega = (3, 3), \quad \omega = (6, 2).$$

The event $X = 7$ is

$$\{\omega \in \Omega : X(\omega) = 7\} = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}.$$

Therefore

$$\mathbb{P}(X = 7) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \in \{7\}\}) = \frac{6}{36} = \frac{1}{6}.$$

The CDF

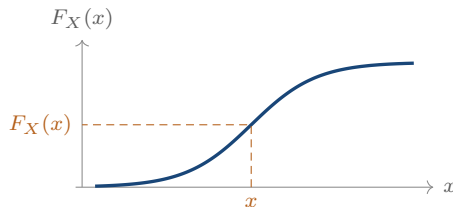
The CDF of any real-valued X is

$$F_X(x) = \mathbb{P}(X \leq x).$$

It satisfies

- F_X is non-decreasing and right-continuous.
- $F_X(-\infty) = 0$ and $F_X(\infty) = 1$.
- For $a < b$,

$$\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a).$$



Probability as area

If the CDF is differentiable, define the **probability density function** by

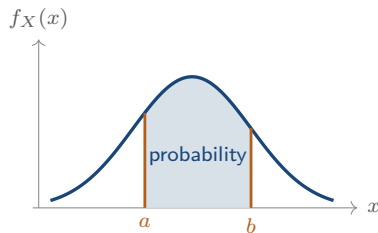
$$f_X(x) := \frac{d}{dx} F_X(x).$$

By the **Fundamental Theorem of Calculus**

$$F_X(x) = \int_{-\infty}^x f_X(t) dt.$$

So for $a < b$,

$$\begin{aligned} \mathbb{P}(a < X \leq b) &= F_X(b) - F_X(a) \\ &= \int_{-\infty}^b f_X(t) dt - \int_{-\infty}^a f_X(t) dt \\ &= \int_a^b f_X(t) dt. \end{aligned}$$



Moments

Continuous expectation

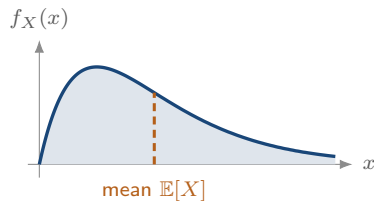
- The **expectation** is the average value.
- If X has density f_X ,

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx.$$

- The **law of the unconscious statistician** (LOTUS) states that

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx.$$

- Each value is weighted by its density, and integration adds the contributions.
- A **moment** is the expectation of a function of a random variable, $\mathbb{E}[g(X)]$.



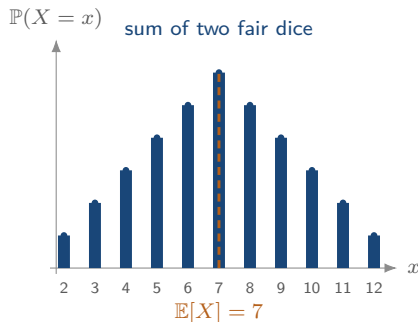
Discrete expectation

If X takes values $x_1, \dots, x_k$, its expectation is the probability-weighted average

$$\mathbb{E}[X] = \sum_{j=1}^k x_j \mathbb{P}(X = x_j).$$

In the discrete case, LOTUS states that

$$\mathbb{E}[g(X)] = \sum_{j=1}^k g(x_j) \mathbb{P}(X = x_j).$$



The Riemann–Stieltjes and Lebesgue definitions unite the discrete and continuous cases.

non-examinable; see appendix

Dummy (indicator) variables

For an event A , define

$$I_A = \begin{cases} 1, & \text{if } A \text{ occurs,} \\ 0, & \text{if } A \text{ does not occur.} \end{cases}$$

By construction,

$$\{I_A = 1\} = A, \quad \{I_A = 0\} = A^c,$$

$$\mathbb{P}(I_A = 1) = \mathbb{P}(A),$$

$$\mathbb{P}(I_A = 0) = \mathbb{P}(A^c).$$

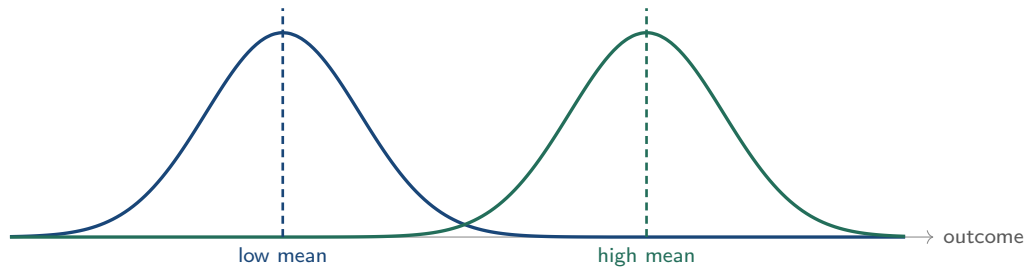
Applying the discrete expectation definition gives

$$\begin{aligned} \mathbb{E}[I_A] &= \sum_{i \in \{0,1\}} i \times \mathbb{P}(I_A = i) \\ &= 0 \times \mathbb{P}(I_A = 0) + 1 \times \mathbb{P}(I_A = 1) \\ &= 0 \times \mathbb{P}(A^c) + 1 \times \mathbb{P}(A) \\ &= \mathbb{P}(A). \end{aligned}$$

Products represent “and”. Since 0 times anything is 0, while $1 \times 1 = 1$, the product $I_A I_B$ equals one exactly when A and B both occur, and zero otherwise. Hence

$$I_A I_B = I_{A \cap B}, \quad \mathbb{E}[I_A I_B] = \mathbb{P}(A \cap B).$$

High and low expectations



Why expectation is linear

Expectations are probability-weighted integrals
(sums in the discrete case).

Integrals and sums are linear.

So, for constants a, b and variables with finite expectations,

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

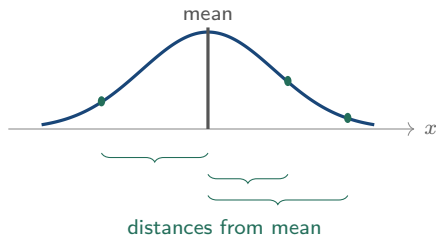
No independence assumption is needed.

Variance

- The **variance** measures average squared distance from the mean:

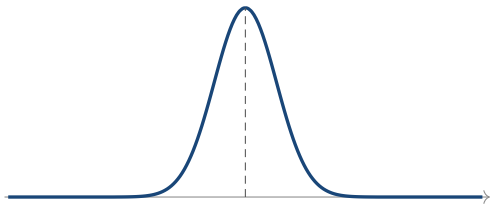
$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2.$$

- A large variance means the distribution is spread out.
- The square matters: positive and negative deviations should not cancel.

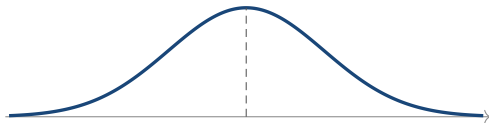


High and low variance

Low variance



High variance



Rescaling spread

Starting from

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2,$$

we have

$$\begin{aligned}\text{Var}(aX) &= \mathbb{E}[(aX)^2] - \mathbb{E}[aX]^2 \\ &= a^2 \mathbb{E}[X^2] - a^2 \mathbb{E}[X]^2 \\ &= a^2 \text{Var}(X).\end{aligned}$$

The **standard deviation** is defined by

$$\text{sd}(X) = \sqrt{\text{Var}(X)}, \quad \text{sd}(aX) = |a| \text{sd}(X).$$

Aside: a standard error is an **estimated** standard deviation: it estimates the standard deviation of an estimator's sampling distribution.

Miles to kilometres

Suppose X is a distance measured in miles. The same distance in kilometres is

$$Y = 1.609X.$$

Variance

$$\text{Var}(Y) = 1.609^2 \text{Var}(X)$$

If $\text{Var}(X)$ is in miles^2 , then $\text{Var}(Y)$ is in km^2 .

Standard deviation

$$\text{sd}(Y) = 1.609 \text{sd}(X)$$

If $\text{sd}(X)$ is in miles, then $\text{sd}(Y)$ is in kilometres.

Variance: squared units

Standard deviation: same units as the variable.

Quantiles and thresholds

For $0 < p < 1$, a general definition of the p th population quantile is

$$q_p = \inf\{x : F_X(x) \geq p\}.$$

This definition also handles discrete outcomes and flat parts of a CDF.

The **median** is the 0.5 quantile:

$$m = q_{0.5}, \quad \mathbb{P}(X \leq m) \geq \frac{1}{2}, \quad \mathbb{P}(X \geq m) \geq \frac{1}{2}.$$

Suppose hourly wages have

$$q_{0.5} = \pounds 18, \quad q_{0.9} = \pounds 35.$$

- Half of wages are at or below $\pounds 18$.
- The top 10% begins at about $\pounds 35$.

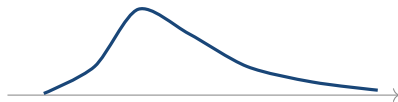
Skewness

- Skewness measures asymmetry:

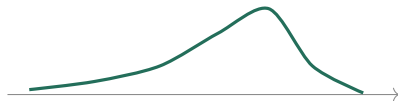
$$\text{Skew}(X) = \mathbb{E} \left[\left(\frac{X - \mu}{\sigma} \right)^3 \right].$$

where $\mu = \mathbb{E}[X]$ and $\sigma = \text{sd}(X)$.

- Symmetric: mean = median.
- Right-skewed: usually mean $>$ median.
- Left-skewed: usually mean $<$ median.



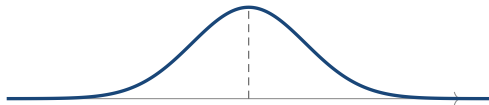
right-skewed



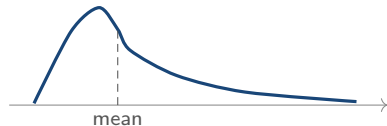
left-skewed

High and low skewness

Low skewness

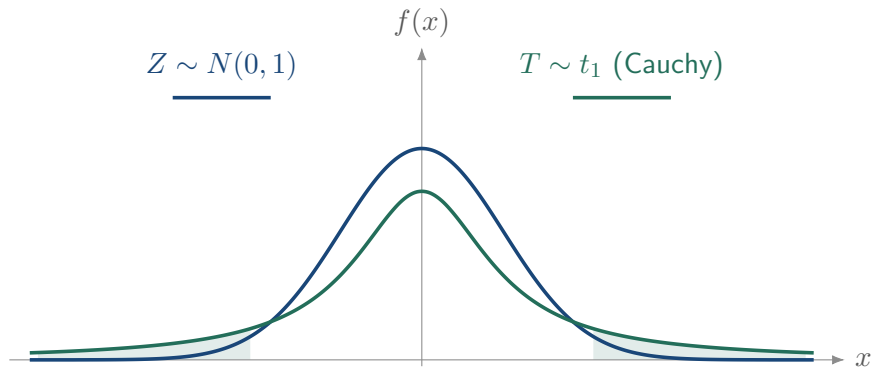


High positive skewness



Examples: income and wealth distributions often have long right tails.

Kurtosis



$$\text{Kurt}(X) = \mathbb{E} \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right].$$

More on Distributions

The uniform distribution

- X is **uniform** on $[a, b]$ if every interval of the same length is equally likely.

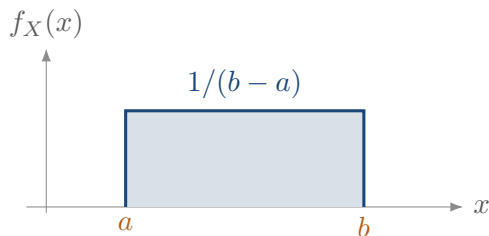
$$X \sim U[a, b].$$

- Its density is flat:

$$f_X(x) = \frac{1}{b-a}, \quad a \leq x \leq b.$$

- The area under the rectangle is one.

$$\mathbb{E}[X] = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}.$$

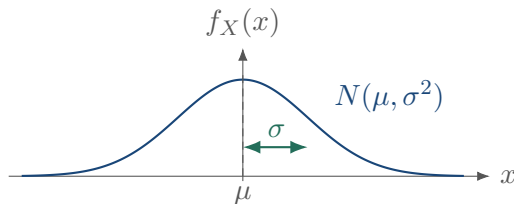


The normal distribution

- Write $X \sim N(\mu, \sigma^2)$, with density

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right].$$

- $\mathbb{E}[X] = \mu$ and $\text{Var}(X) = \sigma^2$.
- The standard normal is $Z \sim N(0, 1)$.
- Its kurtosis is 3, so its **excess kurtosis** is 0.



Can we interpret a density as $\mathbb{P}(X = x)$?

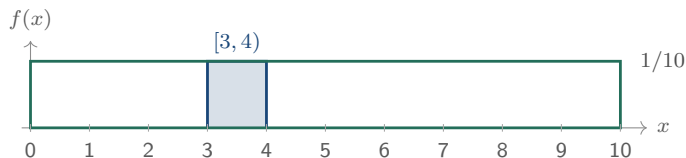
No.

For a continuous random variable,

$$\mathbb{P}(X = x) = 0 \quad \text{even when} \quad f_X(x) > 0.$$

An example: if $X \sim U[0, 10]$, what is $\mathbb{P}(X = \pi)$?

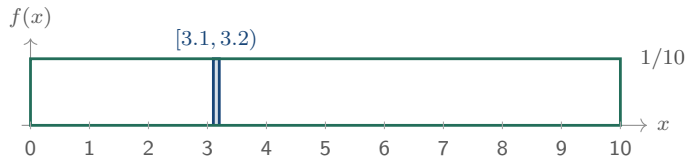
Matching the first digit



Getting the first digit of π right means $X \in [3, 4)$.

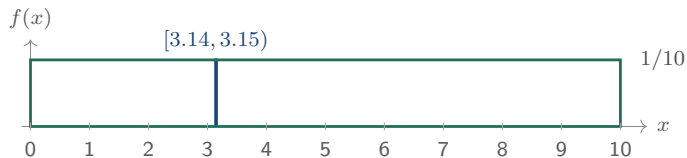
$$\mathbb{P}(3 \leq X < 4) = \frac{\text{width}}{10} = \frac{1}{10}.$$

Matching 3.1



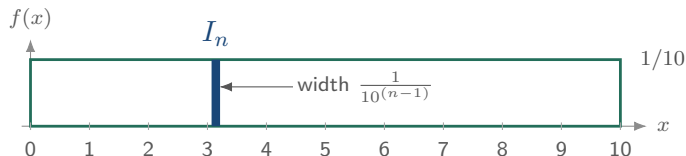
$$\mathbb{P}(3.1 \leq X < 3.2) = \frac{0.1}{10} = \frac{1}{100}.$$

Matching 3.14



$$\mathbb{P}(3.14 \leq X < 3.15) = \frac{0.01}{10} = \frac{1}{10^3}.$$

Getting π



$$\mathbb{P}(\text{first } n \text{ digits right}) = \mathbb{P}(X \in I_n) = \frac{10^{-(n-1)}}{10} = \frac{1}{10^n}.$$

Therefore

$$\mathbb{P}(X = \pi) = \lim_{n \rightarrow \infty} \frac{1}{10^n} = 0, \quad \text{but } f_X(\pi) = \frac{1}{10}.$$

A point can have zero probability even though the density is positive.

What is a density?

Recall that the CDF is

$$F_X(x) = \mathbb{P}(X \leq x).$$

A Taylor expansion of the CDF gives

$$F_X(x') = F_X(x'') + f_X(x'')(x' - x'') + \text{small terms}.$$

Fix any point x_i and set $x' = x_i + h/2$ and $x'' = x_i - h/2$. Then for small $h > 0$,

$$F_X\left(x_i + \frac{h}{2}\right) \approx F_X\left(x_i - \frac{h}{2}\right) + f_X(x_i)h.$$

so

$$\mathbb{P}\left(x_i - \frac{h}{2} < X \leq x_i + \frac{h}{2}\right) \approx f_X(x_i)h.$$

Interpreting density

For equal narrow intervals

$$I_i = \left[x_i - \frac{h}{2}, x_i + \frac{h}{2} \right], \quad i = 1, 2,$$

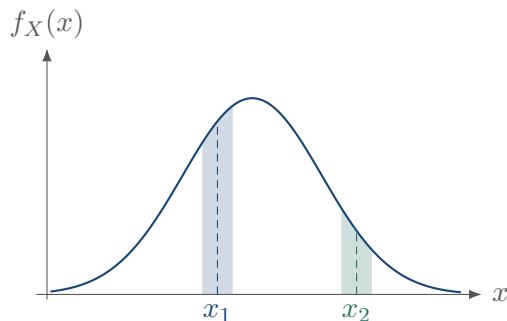
we have

$$\mathbb{P}(X \in I_i) \approx f_X(x_i)h.$$

The common width cancels:

$$\frac{\mathbb{P}(X \in I_1)}{\mathbb{P}(X \in I_2)} \approx \frac{f_X(x_1)}{f_X(x_2)}.$$

Density ratios compare local probabilities.



Not every distribution has a density

- Discrete variables put positive probability on points.
- Continuous variables often have densities.
- Some distributions are mixed or more unusual.
- A compact general notation is

$$\mathbb{E}[g(X)] = \int g(x) \, dF_X(x),$$

which becomes a sum or an ordinary integral in familiar cases.



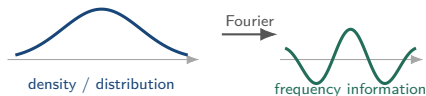
- The characteristic function of X is

$$\varphi_X(t) = \mathbb{E}[e^{itX}].$$

- If X has density f_X , this can be written as

$$\varphi_X(t) = \int_{-\infty}^{\infty} e^{itx} f_X(x) dx.$$

- This is essentially a Fourier transform of the distribution.



Used e.g. to prove the classical CLT

Transformations of Random Variables

Why transform a random variable?

A transformation describes the same underlying outcomes on a more useful scale:

$$Y = g(X).$$

In statistics and econometrics, transformations help us to

- choose interpretable units;
- compare variables on a common scale, for example
- use $\log x$ to compress a long right tail and interpret changes proportionally;
- turn an event into an outcome with $g(x) = \mathbf{1}\{x > c\}$.

$$g(x) = \frac{x - \mu}{\sigma};$$

Once we transform X , we need to find the distribution of Y .

The CDF method

The distribution of $Y = g(X)$ follows from its cumulative probabilities:

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(g(X) \leq y).$$

When g is strictly increasing,

$$\mathbb{P}(g(X) \leq y) = \mathbb{P}(X \leq g^{-1}(y)) = F_X(g^{-1}(y)).$$

Differentiating with respect to y gives the density transformation:

$$f_Y(y) = f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y).$$

Decreasing transformations

If g is strictly decreasing, the inequality reverses:

$$F_Y(y) = \mathbb{P}(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y)).$$

Differentiating, then combining the increasing and decreasing cases, gives the following

For strictly monotonic g :

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

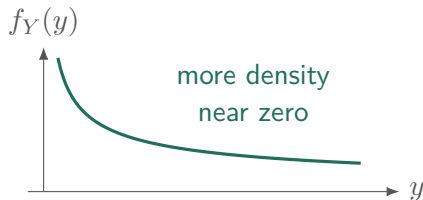
Squaring a uniform variable

Let $X \sim U[0, 1]$ and define $Y = X^2$. For $0 \leq y \leq 1$,

$$F_Y(y) = \mathbb{P}(X^2 \leq y) = \mathbb{P}(X \leq \sqrt{y}) = \sqrt{y},$$

so

$$f_Y(y) = \frac{1}{2\sqrt{y}}, \quad 0 < y < 1.$$



LOTUS by substitution non-examinable

Let $Y = g(X)$, with g differentiable and strictly increasing. The density formula gives

$$f_Y(y) = f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y).$$

Use the substitution

$$u = g^{-1}(y), \quad \frac{du}{dy} = \frac{d}{dy} g^{-1}(y), \quad y = g(u).$$

Then

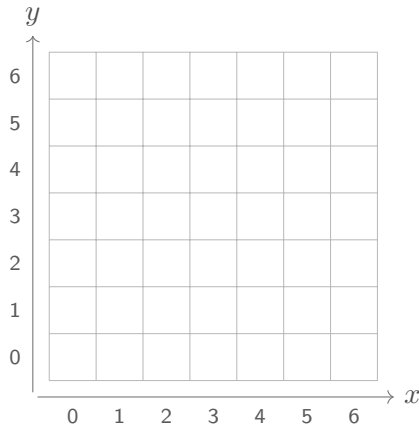
$$\begin{aligned} \mathbb{E}[g(X)] &= \mathbb{E}[Y] \\ &= \int \underbrace{y}_{g(u)} \underbrace{f_X(g^{-1}(y))}_{f_X(u)} \underbrace{\left(\frac{d}{dy} g^{-1}(y)\right) dy}_{du} \\ &= \int g(u) f_X(u) du. \end{aligned}$$

More generally, LOTUS holds for measurable g whenever $\mathbb{E}[|g(X)|] < \infty$.

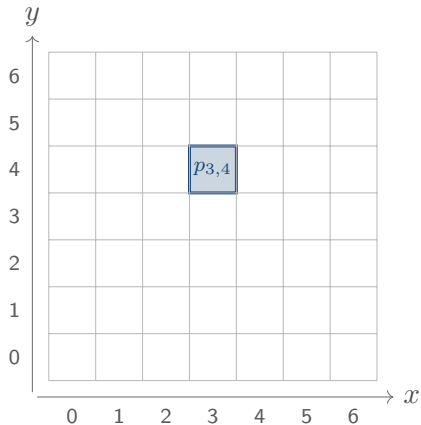
Joint and Conditional Distributions

A joint probability table

- Let X and Y each take values $0, 1, \dots, 6$.
- Each square is one possible pair (x, y) .
- We will use the table as a picture of a joint distribution.



One cell, one probability



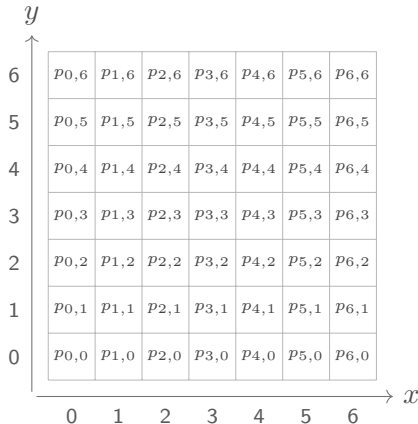
- The highlighted square is the event

$$X = 3, \quad Y = 4.$$

- Its probability is

$$p_{3,4} = \mathbb{P}(X = 3, Y = 4).$$

The joint distribution



- The joint distribution is the full collection

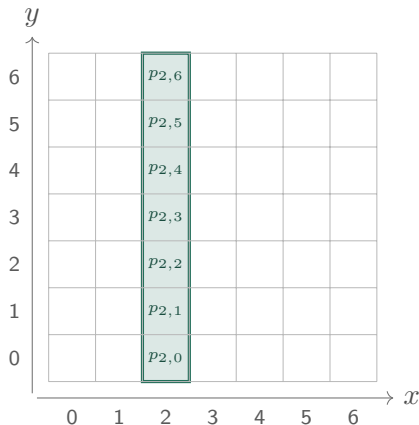
$$\{p_{x,y} : x = 0, \dots, 6; y = 0, \dots, 6\}.$$

- The probabilities must add to one:

$$\sum_{x=0}^6 \sum_{y=0}^6 p_{x,y} = 1.$$

- It tells us how X and Y move together.

Conditioning on $X = 2$

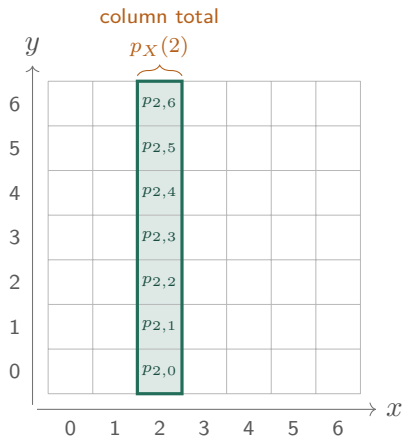


- Focus on the column $X = 2$.
- The conditional distribution of Y given $X = 2$ is the column after normalizing it to add to one:

$$\mathbb{P}(Y = y \mid X = 2) = \frac{p_{2,y}}{\sum_{r=0}^6 p_{2,r}}.$$

- Conditioning means: restrict attention, then renormalize.

The conditional denominator

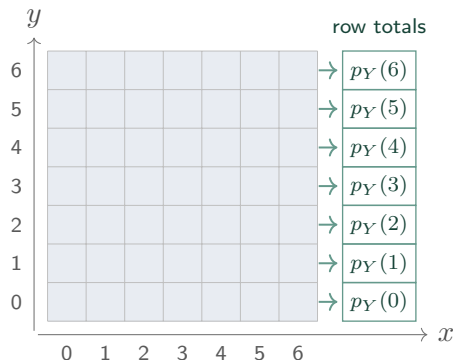


$$p_X(2) = \mathbb{P}(X = 2) = \sum_{r=0}^6 p_{2,r}.$$

- The numerator is one cell.
- The denominator is all probability in the column.
- This is why conditional probabilities add to one over y :

$$\sum_{y=0}^6 \mathbb{P}(Y = y \mid X = 2) = 1.$$

Marginals: row totals

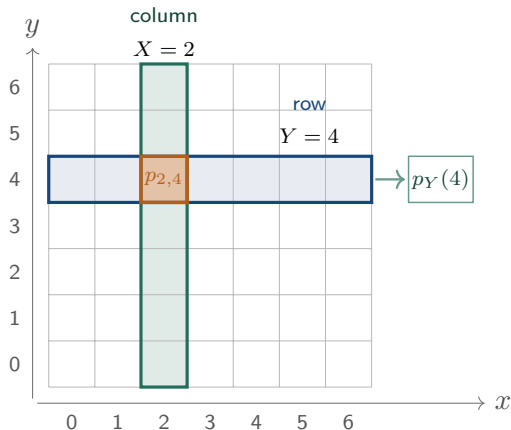


- The marginal distribution of Y ignores X .
- For each row,

$$p_Y(y) = \mathbb{P}(Y = y) = \sum_{x=0}^6 p_{x,y}.$$

- Marginal means “on its own”, after summing out the other variable.

Rows, columns and cells



- A **cell** is one joint probability:

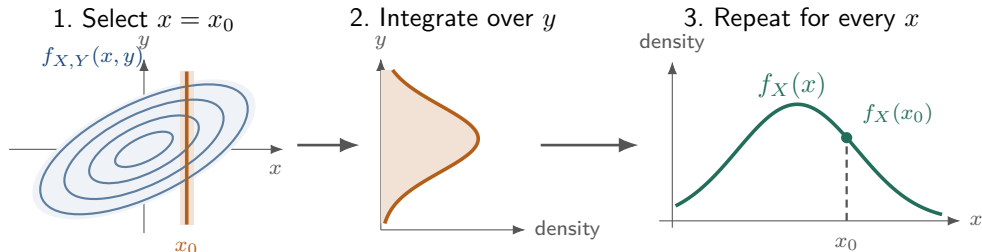
$$p_{2,4} = \mathbb{P}(X = 2, Y = 4).$$

- A **column slice** becomes a conditional distribution after renormalizing.
- A **row total** is a marginal probability:

$$p_Y(4) = \sum_x p_{x,4}.$$

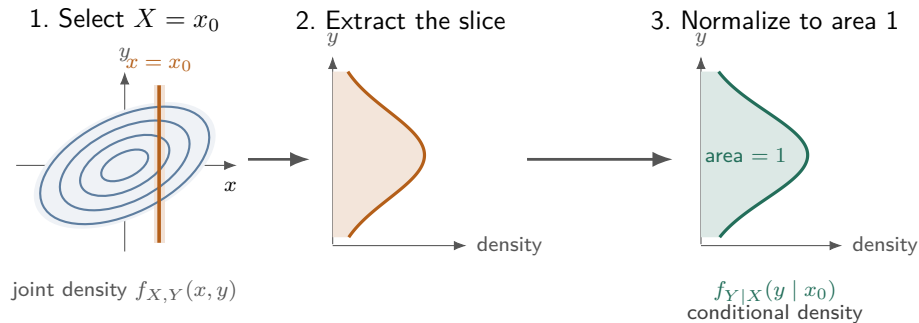
Marginalising

Fix $x = x_0$ and integrate the joint density over y .



$$f_X(x_0) = \int_{\mathbb{R}} f_{X,Y}(x_0, y) \, dy.$$

Conditioning



$$f_{Y|X}(y|x_0) = \frac{f_{X,Y}(x_0,y)}{f_X(x_0)}$$

Bayes rule in x, y notation

- Start from two ways to write the same joint probability:

$$p_{x,y} = p_{Y|X}(y | x)p_X(x) = p_{X|Y}(x | y)p_Y(y).$$

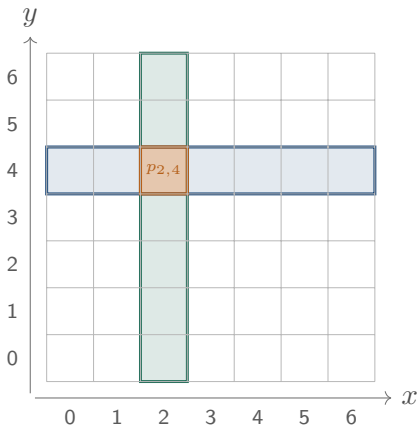
- Rearranging gives Bayes rule:

$$p_{X|Y}(x | y) = \frac{p_{Y|X}(y | x)p_X(x)}{p_Y(y)}.$$

- The denominator is just the marginal probability of $Y = y$:

$$p_Y(y) = \sum_{x'} p_{Y|X}(y | x')p_X(x').$$

Bayes rule in the grid



- The orange cell is the same probability whichever route we use.

$$\mathbb{P}(X = 2, Y = 4)$$

- Column route:

$$\mathbb{P}(Y = 4 \mid X = 2)\mathbb{P}(X = 2).$$

- Row route:

$$\mathbb{P}(X = 2 \mid Y = 4)\mathbb{P}(Y = 4).$$

Bayes rule as updating

- $p_X(x)$ is the **prior** probability of $X = x$.
- $p_{Y|X}(y | x)$ is the **likelihood**: how plausible the evidence is under x .
- After observing y ,

$$p_{X|Y}(x | y) \propto p_{Y|X}(y | x)p_X(x).$$

- The denominator rescales so probabilities add to one.

$$\text{posterior} = \frac{\text{likelihood} \times \text{prior}}{\text{normalising constant}}$$

$$p_{X|Y}(x | y) = \frac{p_{Y|X}(y | x)p_X(x)}{\sum_{x'} p_{Y|X}(y | x')p_X(x')}.$$

Bayes rule for events A and B

For any events A and B with positive probability,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}.$$

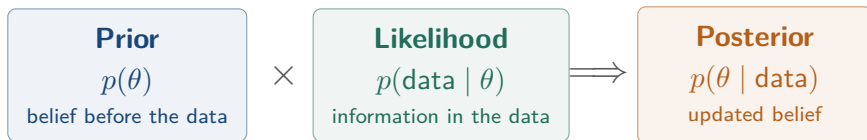
If A and A^c are the two possible states, the probability of the evidence is

$$\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A) + \mathbb{P}(B \mid A^c)\mathbb{P}(A^c).$$

$\mathbb{P}(A)$	$\mathbb{P}(B \mid A)$	$\mathbb{P}(A \mid B)$
prior belief	evidence under A	updated belief

Lecture 1: Bayesian inference

Bayesian probability represents a **strength of belief**; observing data updates that belief using Bayes rule.



$$p(\theta \mid \text{data}) = \frac{p(\text{data} \mid \theta) p(\theta)}{p(\text{data})}.$$

Posterior $\propto$ likelihood $\times$ prior.

A signal about a hidden state

Let H mean that a borrower will default, and let S be a risk signal.

$$\mathbb{P}(H) = 0.10,$$

$$\mathbb{P}(S | H) = 0.80, \quad \mathbb{P}(S | H^c) = 0.20.$$

We observe the signal S .

First find the probability of the evidence:

$$\mathbb{P}(S) = 0.80(0.10) + 0.20(0.90) = 0.26.$$

Then update:

$$\mathbb{P}(H | S) = \frac{0.80(0.10)}{0.26} = \frac{4}{13} \simeq 0.31.$$

Our belief that the borrower would default was 0.10; after observing the signal, it is now 0.31.

Bayes rule in odds form

$$\underbrace{\frac{\mathbb{P}(A | B)}{\mathbb{P}(A^c | B)}}_{\text{posterior odds}} = \underbrace{\frac{\mathbb{P}(A)}{\mathbb{P}(A^c)}}_{\text{prior odds}} \times \underbrace{\frac{\mathbb{P}(B | A)}{\mathbb{P}(B | A^c)}}_{\text{likelihood ratio}}.$$

> 1

B is more likely under A .
The posterior odds of A
rise.

$= 1$

B is equally likely under
either state.
The odds do not change.

< 1

B is more likely under A^c .
The posterior odds of A
fall.

Posterior odds combine prior beliefs with the strength of the evidence.

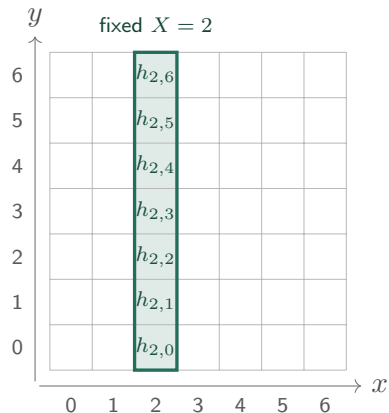
Conditional Moments

Conditional expectation

- For a function $h(X, Y)$,

$$\begin{aligned}\mathbb{E}[h(X, Y) \mid X = x] \\ = \sum_y h(x, y) p_{Y|X}(y \mid x).\end{aligned}$$

- Hold x fixed, then average over the remaining uncertainty in Y .
- The weights are the conditional probabilities in that column.



Conditional means in a table

fixed $X = 2$

4			$h_{2,4}$		
3			$h_{2,3}$		
2			$h_{2,2}$		
1			$h_{2,1}$		
0			$h_{2,0}$		
	0	1	2	3	4

$$\mathbb{E}[h(X, Y) \mid X = 2] = \sum_{y=0}^4 h_{2,y} p_{Y|X}(y \mid 2).$$

- Values: $h_{2,y}$.
- Weights: conditional probabilities.

Conditional variance

- Conditional variance measures spread after conditioning:

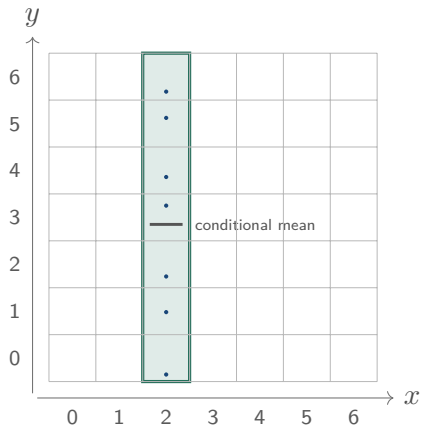
$$\text{Var}(h(X, Y) \mid X = x) = \mathbb{E} \left[(h(X, Y) - \mathbb{E}[h(X, Y) \mid X = x])^2 \mid X = x \right].$$

- Equivalent computational form:

$$\text{Var}(h(X, Y) \mid X = x) = \mathbb{E}[h(X, Y)^2 \mid X = x] - \left(\mathbb{E}[h(X, Y) \mid X = x] \right)^2.$$

- It is the variance within one column of the grid, using conditional weights.

Conditional variance in a table



- In a column, the conditional mean is the centre.
- Conditional variance asks how far the column's values sit from that centre.
- We can condition on $Y = y$ instead and do the same calculation along a row.

Two Laws of Conditioning

Counting the whole grid

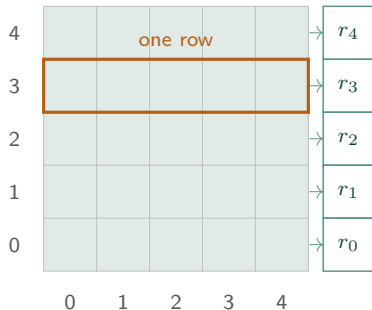
	y				
4	$h_{0,4}$	$h_{1,4}$	$h_{2,4}$	$h_{3,4}$	$h_{4,4}$
3	$h_{0,3}$	$h_{1,3}$	$h_{2,3}$	$h_{3,3}$	$h_{4,3}$
2	$h_{0,2}$	$h_{1,2}$	$h_{2,2}$	$h_{3,2}$	$h_{4,2}$
1	$h_{0,1}$	$h_{1,1}$	$h_{2,1}$	$h_{3,1}$	$h_{4,1}$
0	$h_{0,0}$	$h_{1,0}$	$h_{2,0}$	$h_{3,0}$	$h_{4,0}$
	0	1	2	3	4
	x				

- Start with an equally weighted grid.
- Add every cell once:

$$T = \sum_x \sum_y h(x, y).$$

- Next: count the same T by rows or by columns.

Counting by row totals

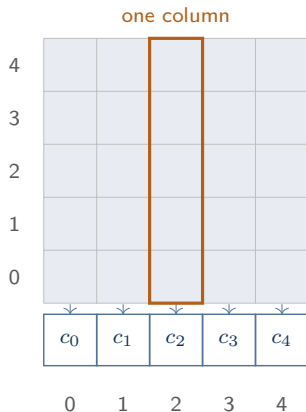


$$r_y = \sum_x h(x, y)$$

$$T = \sum_y r_y = \sum_y \sum_x h(x, y).$$

- Add across each row.
- Then add the row totals.

Counting by column totals



$$c_x = \sum_y h(x, y)$$

$$T = \sum_x c_x = \sum_x \sum_y h(x, y).$$

- Add down each column.
- Then add the column totals.

From totals to averages

- If the n^2 cells are equally weighted, the overall average is

$$\frac{1}{n^2} \sum_x \sum_y h(x, y).$$

- Using row averages first gives

$$\frac{1}{n} \sum_y \left\{ \frac{1}{n} \sum_x h(x, y) \right\}.$$

- Using column averages first gives

$$\frac{1}{n} \sum_x \left\{ \frac{1}{n} \sum_y h(x, y) \right\}.$$

Average all at once, or **average inside groups**
and then average the group averages.

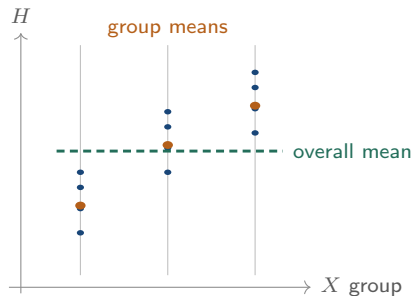
The law of iterated expectations

For $H = h(X, Y)$, the law of iterated expectations is

$$\mathbb{E}[H] = \mathbb{E}[\mathbb{E}[H \mid X]] = \mathbb{E}[\mathbb{E}[H \mid Y]].$$

Average within values of the conditioning variable, then average those conditional means across groups.

Total variance: intuition



- Total variation has two parts.
- **Within-group variation**: how spread out H is inside each value of X .
- **Between-group variation**: how much the conditional means differ across X .

The law of total variance

$$\text{Var}(H) = \mathbb{E}[\text{Var}(H \mid X)] + \text{Var}(\mathbb{E}[H \mid X]).$$

A self-contained proof is in the appendix [non-examinable](#).

Independence Concepts

Independence of events

- $\mathbb{P}(A)$: how likely I think A is to happen.
- $\mathbb{P}(A \mid B)$: after learning that B happened, how likely I now think A is to happen.

If you knew that B occurred, it would not change your opinion about A .

When $\mathbb{P}(B) > 0$, this means

$$\mathbb{P}(A \mid B) = \mathbb{P}(A).$$

Using $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$, this is equivalent to

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Conditional independence

$$A \perp\!\!\!\perp B \mid W$$

Once we know W , learning B gives no further information about A .

$$\begin{aligned} &\text{For each state } w \text{ with } \mathbb{P}(W = w) > 0, \\ &\mathbb{P}(A \cap B \mid W = w) = \mathbb{P}(A \mid W = w)\mathbb{P}(B \mid W = w). \end{aligned}$$

Example. Two people decide independently whether to carry umbrellas, given the weather. Both are more likely to carry one on rainy days.

Independence within each group need not imply independence overall.

Independence of random variables

Knowing the value of X does not change the distribution of Y .

Formally, X and Y are independent if, for every pair of sets $C, D \subseteq \mathbb{R}$,

$$\mathbb{P}(X \in C, Y \in D) = \mathbb{P}(X \in C)\mathbb{P}(Y \in D).$$

Independent and identically distributed (iid). A collection $X_1, \dots, X_n$ is iid when

- the variables are independent across observations i ; and
- every X_i has the same distribution.

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F.$$

Mean independence

Knowing X does not change the mean you predict for Y .

Y is mean independent of X if

$$\mathbb{E}[Y \mid X] = \mathbb{E}[Y].$$

What must stay the same

The conditional mean of Y .

What may still change

The variance, tails and shape of $Y \mid X$.

Mean independence is **directional** and weaker than independence: it fixes only the mean.

Conditional transformations

Suppose outcomes differ in both location and spread across groups:

$$Y \mid X = x \sim N\left(\mu(x), \sigma^2(x)\right).$$

Standardise within each group:

$$Z = \frac{Y - \mu(X)}{\sigma(X)}.$$

Then, for every value of x ,

$$Z \mid X = x \sim N(0, 1).$$

The entire conditional distribution is now constant in x , so $Z \perp\!\!\!\perp X$;
equal conditional means alone would not be enough.

Covariance and correlation

Covariance measures how two variables move together:

$$\text{Cov}(A, B) = \mathbb{E}[(A - \mathbb{E}[A])(B - \mathbb{E}[B])] = \mathbb{E}[AB] - \mathbb{E}[A]\mathbb{E}[B].$$

Correlation removes the units:

$$\text{Corr}(A, B) = \frac{\text{Cov}(A, B)}{\text{sd}(A) \text{sd}(B)} \in [-1, 1].$$

Positive: move together. Negative: move in opposite directions.

Assume both variances are finite and positive.

Uncorrelatedness

X and Y have no linear relationship.

X and Y are **uncorrelated** if

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = 0.$$

What must be zero

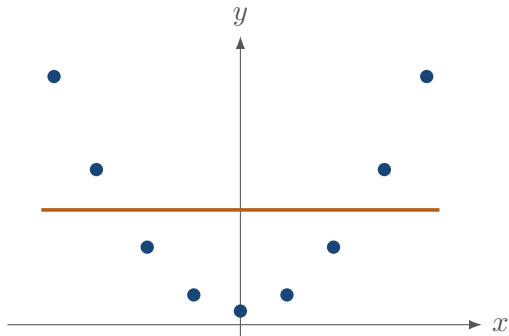
One number: the covariance.

What may still remain

A nonlinear relationship, including a changing conditional mean.

Unlike mean independence, uncorrelatedness does **not** require $\mathbb{E}[Y \mid X] = \mathbb{E}[Y]$.

Uncorrelatedness is linear



Preview of Lecture 3: OLS in

$$y_i = \beta_0 + \beta_1 x_i + u_i,$$

gives

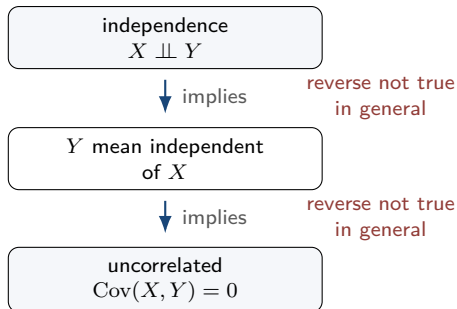
$$\hat{\beta}_1 = \frac{\widehat{\text{Cov}}(x, y)}{\widehat{\text{Var}}(x)}.$$

We can see that

$$\hat{\beta}_1 = 0 \quad \implies \quad \widehat{\text{Cov}}(x, y) = 0$$

Even though knowing x would change your opinion about the mean of y

Implications



- Independence concerns the whole distribution.
- Mean independence only fixes the conditional mean.
- Uncorrelatedness only fixes one covariance.

Mean independence without independence

Let $X \sim \text{Bernoulli}(1/2)$ and let $U \in \{-1, 1\}$ be independent of X , with $\mathbb{P}(U = 1) = \mathbb{P}(U = -1) = 1/2$. Define $Y = XU$.

Mean independence holds

$$\mathbb{E}[Y \mid X = 0] = 0,$$

$$\begin{aligned}\mathbb{E}[Y \mid X = 1] &= \mathbb{E}[U \mid X = 1] \\ &= \mathbb{E}[U] = 0.\end{aligned}$$

Hence

$$\mathbb{E}[Y \mid X] = 0 = \mathbb{E}[Y].$$

Independence fails

Because U is never zero, $Y = 0$ exactly when $X = 0$. Therefore

$$\mathbb{P}(Y = 0) = \mathbb{P}(X = 0) = \frac{1}{2},$$

whereas

$$\mathbb{P}(Y = 0 \mid X = 0) = 1.$$

Note: $\text{Var}(Y \mid X = 0) = 0$, while $\text{Var}(Y \mid X = 1) = \text{Var}(U) = 1$:
the conditional variance changes with X .

Uncorrelated but mean dependent

Let $X \in \{-1, 0, 1\}$ with equal probabilities, and define $Y = X^2$.

Uncorrelated

$$\mathbb{E}[X] = \frac{-1 + 0 + 1}{3} = 0,$$

$$\mathbb{E}[Y] = \frac{1 + 0 + 1}{3} = \frac{2}{3},$$

$$\mathbb{E}[XY] = \mathbb{E}[X^3] = \frac{-1 + 0 + 1}{3} = 0.$$

and so $\text{Cov}(X, Y) = 0 - (0)(2/3) = 0$.

Not mean independent

Since $Y = X^2$,

$$\mathbb{E}[Y \mid X = x] = x^2.$$

In particular,

$$\mathbb{E}[Y \mid X = 0] = 0 \neq \frac{2}{3} = \mathbb{E}[Y].$$

Y and X have a **non-linear** relationship

Variance of a sum

For random variables A and B with finite variances,

$$\text{Var}(A + B) = \text{Var}(A) + \text{Var}(B) + 2 \text{Cov}(A, B).$$

Independence implies $\text{Cov}(A, B) = 0$.

It is sufficient for variances to add, but **not necessary**.

The important condition is **zero covariance**.

In the previous example, X and X^2 are dependent but uncorrelated:

$$\text{Var}(X + X^2) = \text{Var}(X) + \text{Var}(X^2).$$

OLS: unbiasedness and consistency

For OLS on $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, the headline conditions are:

Unbiasedness

The key condition is that the error is **mean independent** of all the regressors:

$$\mathbb{E}[\varepsilon_i \mid x_1, \dots, x_n] = \mathbb{E}[\varepsilon_i] = 0.$$

Consistency

The key condition is that x_i and ε_i are **uncorrelated**:

$$\text{Cov}(x_i, \varepsilon_i) = 0.$$

When $\mathbb{E}[\varepsilon_i] = 0$, this is often written

$$\mathbb{E}[x_i \varepsilon_i] = 0.$$

In Summary

- Random variables map outcomes to numbers; CDFs, probability masses and densities describe their distributions.
- Expectations, variance, quantiles, skewness and kurtosis summarize location, spread and shape.
- Transformations change scale; the CDF method, density rule and LOTUS tell us how distributions and expectations change.
- Joint, marginal and conditional distributions underpin Bayes' rule, iterated expectations and the law of total variance.
- Independence, mean independence and zero correlation are different concepts, with different implications for regression.

Next time

Lecture 3: Statistics II

Defining expectation non-examinable

Riemann–Stieltjes integral

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) \, dF_X(x).$$

This integrates directly with respect to the CDF. Its smooth increases and jumps supply probability weights.

Lebesgue integral

$$\mathbb{E}[g(X)] = \int_{\Omega} g(X(\omega)) \, d\mathbb{P}(\omega).$$

This integrates over the sample space, weighting outcomes by their probabilities.

Both cover discrete, continuous and mixed distributions. If X has a density, they reduce to LOTUS:

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) \, dx.$$

Total variance: proof non-examinable

Let

$$\mu = \mathbb{E}[H], \quad m(X) = \mathbb{E}[H \mid X].$$

Then $\mathbb{E}[m(X)] = \mu$ and

$$H - \mu = (H - m(X)) + (m(X) - \mu).$$

Squaring and taking expectations gives

$$\begin{aligned} \text{Var}(H) &= \mathbb{E}[(H - m(X))^2] + \text{Var}(m(X)) \\ &\quad + 2\mathbb{E}[(H - m(X))(m(X) - \mu)]. \end{aligned}$$

By the definition of conditional variance and $\mathbb{E}[\mathbb{E}[Z \mid X]] = \mathbb{E}[Z]$, the first term is $\mathbb{E}[\text{Var}(H \mid X)]$. Since $m(X) - \mu$ is determined by X , the cross term is zero because

$$\mathbb{E}[(H - m(X))(m(X) - \mu)] = \mathbb{E}\left[(m(X) - \mu) \underbrace{\mathbb{E}[H - m(X) \mid X]}_{=0}\right] = 0.$$

Therefore

$$\text{Var}(H) = \mathbb{E}[\text{Var}(H \mid X)] + \text{Var}(\mathbb{E}[H \mid X]).$$